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Robust, partially alive particle Metropolis-Hastings via The Frankenfilter

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Abstract

When a hidden Markov model permits the conditional likelihood of an observation given the hidden process to be zero, all particle simulations from one observation time to the next could produce zeros. If so, the filtering distribution cannot be estimated and the estimated parameter likelihood is zero. The Alive Particle Filter addresses this by simulating a random number of particles for each inter-observation interval, stopping after a target number of non-zero conditional likelihoods. For outlying observations or poor parameter values, a non-zero result can be extremely unlikely, and computational costs prohibitive. We introduce the Frankenfilter, a principled, partially alive particle filter that targets a user-defined amount of success whilst fixing lower and upper bounds on the number of simulations. The Frankenfilter produces unbiased estimators of the likelihood, suitable for pseudo-marginal Metropolis–Hastings (PMMH). We demonstrate that PMMH with the Frankenfilter is more robust to outliers and mis-specified initial parameter values than PMMH using standard particle filters, and is typically at least 2-3 times more efficient. We also provide advice for choosing the amount of success. In the case of n exact observations, this is particularly simple: target n successes.

Keywords: particle MCMC, Markov jump process, particle filter, alive particle filter, unbiased estimator.

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1 Introduction

Consider a hidden Markov model for a d_x-dimensional hidden process, $\{X_t\}_{t\in\mathcal{T}}$, with index set $\mathcal{T}\subseteq[0,T]$ and state space $\mathcal{X}$; d_y-dimensional observations $y_i\in\mathcal{Y}$, $i=1,\ldots,n$, of the process are made at times $t_i\in\mathcal{T}$, with $t_n=T$. The likelihood for the parameter, θ, is

$$P(y_{1:n}\mid\theta)=\int f(y_{1:n}\mid x_T,\theta)\mathrm{d}\tilde{P}(x_T\mid\theta),$$

where $\tilde{P}$ is the law of x_T, and the conditional likelihood is

$$f(y_{1:n}\mid x_T,\theta)=\prod_{i=1}^{n} f(y_i\mid x_{t_i},\theta).$$

Suppose that $f(y_i\mid X_{t_i}=x,\theta)=0$ if some criterion involving X_{t_i} and y_i is not satisfied. A case of particular note is when each conditional likelihood is an indicator function: $f(y_i\mid X_{t_i}=x):\mathcal{Y}\times\mathcal{X}\to\{0,1\}$, for example with exact or partial but noiseless observations of a Markov jump process (MJP), which has a discrete state space, $\mathcal{X}\subseteq\mathbb{Z}^d$; alternative scenarios include, for any Markov process, approximate Bayesian computation with a kernel such as $1(\|y_i-x_{t_i}\|\le\epsilon)$ (e.g., Sisson et al., 2018).

The usual collection of particle filters, including the bootstrap filter Gordon et al. (1993) and the auxiliary particle filter Pitt and Shephard (1999) fix the number of particles (simulations) from one observation time, t_{i-1} to the next, t_i. Thus, when $f(y_i\mid X_{t_i}=x)=0$ is possible, the conditional likelihood of every particle could be 0; in this case, none of the particles at time t_i can be resampled, and the particle filter 'dies out'. Consequently, it is not possible to estimate the filtering distribution and, within a particle MCMC or SMC^2 scheme, the estimate of the likelihood is 0.

When a traditional particle filter is used within particle Metropolis–Hastings, the ideal number of particles can vary from one parameter value to another; for example, a parameter value θ that

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fits less well with the data than the posterior mode, $\hat{\theta}$, will lead to simulations from the prior, $\tilde{P}(\cdot|\theta)$, of the hidden process that fit less well with the data, and this can lead to more Monte Carlo variability in the estimate of the log likelihood for that particular simulation, necessitating a larger number of particles. Since in real scenarios (nearly) all models are wrong, even under $\tilde{P}(\cdot|\hat{\theta})$, deviations between the path of the true process (which produced $y_{1:n}$) and probable values under the model, or a mis-specified observation model, could lead to some of the observations being much less likely (under the model) than others, necessitating more simulations for the related inter-observation intervals. All of these points have motivated the development of the Frankenfilter.

Notation and simplification. We suppress dependence on the parameter, θ, from the notation. In Sections 1-3, purely to simplify the presentation, we assume integer observation times ($t_i = i$, $i = 1,\ldots,n$; $t_n = n = T$) and that the transition kernel and observation process are time homogeneous. Here and throughout, $p_0(\cdot)$ denotes the prior for the hidden process at time 0, and the one step transition kernel from time $t-1$ to time t, conditional on $X_{t-1} = x$, is $p(\cdot|x)$. The likelihood for the data conditional on θ becomes $P(y_{1:T})$.

1.1 The alive particle filter and compromises in practice

LeGland and Oudjane (2004, 2005) propose and analyse a particle filter which, for each inter-observation interval, simulates repeatedly until a prespecified amount of 'success', $\mathfrak{s}$, has been achieved, thus producing a random number of particles. In the special case of a product of binary (indicator) likelihoods, simulations are started from a state chosen uniformly at random from the successful states at time t, $\mathfrak{s}$ is the target number of successes (1s) and the estimate of the likelihood for T observations is $\hat{P}_{LO} := \prod_{t=1}^{T} \frac{\mathfrak{s}}{M_t}$, where M_t is the number of simulations required to achieve $\mathfrak{s}$ "successes" at time t.

Whilst $\hat{P}_{LO}$ is consistent for the true likelihood, p, as $\mathfrak{s} \to \infty$, it is biased and is, therefore, unsuitable for particle Metropolis–Hastings. Amrein and Künsch (2011) and del Moral et al.

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(2015) consider the case of a product of indicator function conditional likelihoods, and show that $\mathbb{E}\left[\hat{P}_{\text{alive}}\right] = P(y_{1:T})$, where

$$\hat{P}_{\text{alive}} := \prod_{t=1}^{T} \frac{\mathfrak{s}-1}{M_t - 1}.$$

Clearly, we must set $\mathfrak{s} \geq 2$. Subject to strong conditions, del Moral et al. (2015) also shows that to control the variance of $\hat{P}_{\text{alive}} / p$, $\mathfrak{s}$ should grow linearly with *T*.

Each M_t is now unbounded and can be exceedingly large when the success probability is small, leading to a large, and potentially even infinite, expected runtime of the algorithm. The current ad hoc solution to this problem Drovandi and McCutchan (2016); Drovandi et al. (2016) fixes a (high) hard threshold $m_+ \geq \mathfrak{s}$ on M_t such that if this threshold is reached, $M_t = m_+$, simulation halts and $\hat{P}_{\text{alive}} = 0$ is returned. The motivation is that if $M_t = m_+ >> \mathfrak{s}$, the simulated process does not fit with the data, so the associated parameter value should have low posterior mass and, hence, should be rejected.

The 'alive particle filter with a hard threshold', where success/failure is binary (as in del Moral et al. (2015); Drovandi and McCutchan (2016); Drovandi et al. (2016)) is detailed in Algorithm 1. Whether the Markov process is discrete in time or continuous in time, we refer to the set of states needed to propagate from x_{t-1} to (and including) x_t as the *path* $x_{(t-1,t]}$ (strictly, $x_{(t-1,t]\cap T}$). The algorithm counts a particle at time *t* as successful if $f(y_t \mid x_t) = 1$ and as a failure if $f(y_t \mid x_t) = 0$. It starts by sampling a particle from the prior, p_0, and propagating it to time 1, repeating until $\mathfrak{s}$ time-1 successes have been achieved. For each subsequent time, $t = 2,\ldots,T$, first an *ancestor* is sampled uniformly from the pool of the first $\mathfrak{s}-1$ successful particles at time $t-1$; this particle is then propagated to time *t* and, if successful it is added to the pool of particles that were successful at time *t*; again, this process is repeated until $\mathfrak{s}$ time-*t* successes have been achieved.

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Because of the hard threshold, Algorithm 1 leads to a biased estimator of the likelihood and biased inference for parameters within pseudo-marginal MCMC or SMC^2. The first simulation study, in Section 4.1, demonstrates this bias in practice.

Algorithm 1 Alive Filter with Hard Threshold

1: **procedure** ALIVETHRESH($\mathfrak{s}, m_+$) ▷ target # successes, max #trials

2: **for** *t* in 1: *T* **do**

3: $m \leftarrow 0$.

4: **do**

5: $m \leftarrow m+1$.

6: **if** $t=1$ **then**

7: Sample $x_{t-1}^m \sim p_0$; set $a_{t-1}^m = m$.

8: **else**

9: $a_{t-1}^m \leftarrow$ **DiscreteUniform**($\{1,\ldots,\mathfrak{s}-1\}$).

10: **end if**

11: Sample $\tilde{x}_t^m \sim p(\cdot | x_{t-1}^{a_{t-1}^m})$; calculate $s^m \in \{0,1\}$ from (x_t^m, y_t).

12: **if** $s^m = 1$ **then**

13: $x_t^{\sum_{j=1}^m s^j} = \tilde{x}_t^m$

14: **end if**

15: **while** $m < m_+$ **and** $\sum_{j=1}^m s^j < \mathfrak{s}$

16: **if** $m = m_+$ **then**

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17: **return** $\hat{p} = 0$. ▷ $\hat{p}_t = 0 \Rightarrow \prod_{s=1}^{T} \hat{p}_s = 0$

18: **else**

19: $\hat{p}_t \leftarrow \frac{\mathfrak{s}-1}{m-1}$.

20: **end if**

21: **end for**

22: **return** $\hat{p} = \prod_{s=1}^{T} \hat{p}_s$.

23: **end procedure**

Inspired by Shelley (1818) and motivated by the desire for the robustness and adaptivity of the fully alive filter and the practical necessity of bounding the amount of computational work, we present the Frankenfilter, a partially alive particle filter that creates an unbiased estimator of the likelihood. Although motivated by issues with indicator likelihoods and MJPs, the Frankenfilter may be employed for its robustness and adaptivity across the same set of Markov process and conditional likelihood regimes as the standard particle filter.

1.2 Our contribution and the structure of this article

The Frankenfilter corrects the alive particle filter with a hard threshold so that the likelihood estimator is unbiased, and generalises the applicability in several directions. In particular, the Frankenfilter allows: (i) specification of a minimum number of simulations, m_-, as well as a maximum, m_+; (ii) the conditional likelihood to be any non-negative weight rather than an indicator; in particular, this permits the use of informed transition proposals, so called 'bridges' Golightly and Wilkinson (2015); Golightly and Sherlock (2019); (iii) any non-negative measure of success, rather than an indicator; this permits the amount of success to depend on the conditional likelihood, for example.

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We also derive tuning advice for the alive particle filter and the Frankenfilter. Simulation studies validate the tuning advice and compare PMMH using the Frankenfilter with an equivalent static particle filter, exploring the relative robustness and efficiency.

The Frankenfilter can be applied to any Markov process observed at a discrete set of times, where observations y_t have a known conditional likelihood, $f(y_t \mid X_t = x_t)$, given the state of the hidden process, $X_t = x_t$, $t = 1,\ldots,T$. We will repeatedly refer to two cases of particular interest, where the conditional likelihood is an indicator function:

• *Full, exact observations*: At each observation time, the full state vector is observed; *i.e.*, we observe $X_t = x_t$, $t = 1,\ldots,T$.

• *Partial, exact observations*: At each observation time, a subset of the components of the state vector are observed. Let $X_t = (X_t^*, Z_t)$; we observe $X_t^* = x_t^*$, $t = 1,\ldots,T$.

Section 2 presents the Frankenfilter in several stages of increasing generality and proves the unbiasedness of the likelihood estimator. Section 3 derives advice on tuning the alive particle filter and the Frankenfilter and Section 4 presents the simulation studies. The article concludes in Section 5 with a discussion.

2 The Frankenfilter

To maximise the accessibility of the key ideas, we build up from the simplest possible scenario, dealt with by Algorithm 2, generalise it in several ways (Algorithm 3), proving unbiasedness in the main text, and conclude with the full Frankenfilter (Algorithm 4).

2.1 Full, exact observations

We first present the simplest Frankenfilter, Algorithm 2, which estimates $p = \mathbb{P}_{\tilde{P}}\left(X = x\right)$ for some *x* by repeatedly sampling $X \sim \tilde{P}$ and counting the number of successes ($X = x$). Here, for the *j*th simulation, $w^j := w(x^j) \in \{0,1\}$ indicates failure or success; for our purposes, $w(x^j) = 1(x^j = x)$. We fix the desired number of successes, $\mathfrak{s} \geq 2$, and an upper bound on the

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number of simulations, $m_+ \geq \mathfrak{s}$. The algorithm stops when either the number of successes reaches $\mathfrak{s}$ or m_+ iterations have been performed.

If $\mathfrak{s}=1$ then success could be achieved after the first simulation, leading to an estimate of $0/0$, which is undefined. Strictly, the algorithm is valid, producing an unbiased estimator of the success probability when $\mathfrak{s} \geq 2$. Typically, we will ask for $\mathfrak{s} >> 2$; see Section 3.

In the case of full-vector, exact observations of a Markov jump process and a known initial condition, Algorithm 2 would be applied separately for the initial interval (simulating $X_1 \mid X_0 = x_0$) and then each inter-observation interval (simulating $X_t \mid X_{t-1} = x_{t-1}$).

Algorithm 2 Basic One-step Frankenfilter

1: **procedure** FRANKENFILTERBASEONE($\mathfrak{s}, m_+$) ▷ target # successes, max trials

2: $m \leftarrow 0$.

3: **do**

4: $m \leftarrow m+1$.

5: Sample x^m from $\tilde{P}$; calculate associated weight, $w^m \in \{0,1\}$.

6: **while** $m < m_+$ **and** $\sum_{j=1}^{m} w^j < \mathfrak{s}$

7: **if** $\sum_{j=1}^{m} w^j < \mathfrak{s}$ **then return** $\hat{p} = \frac{1}{m}\sum_{j=1}^{m} w^j$.

8: **else return** $\hat{p} = \frac{1}{m-1}\sum_{j=1}^{m-1} w^j$.

9: **end if**

10: **end procedure**

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Algorithm 3 generalises Algorithm 2, by allowing a pre-specified minimum number of simulations, for importance-based proposals and for an arbitrary (non-binary) measure of success. It also separates the measure of success from the particle weight.

As with Algorithm 2 it can be applied in the case of full-vector, exact observations, with realisations of independent estimators obtained for each inter-observation interval. The proof that the estimator produced by Algorithm 3 is unbiased is included in the main text as the intuition is helpful for understanding the structure of the estimator.

Pathak (1976) describes an unbiased estimator when samples of a real-valued variate are taken sequentially and each sample has an associated cost, which can only be ascertained after the sample has been taken. Sampling stops when a specific cost budget is reached. Kremers (1987) considers Bernoulli variables and assumes a fixed, constant cost for samples, but allows upper and lower bounds on the number of samples taken. Combining these ideas, we sample $X^j \sim Q$, for some proposal distribution, *Q*, with density/mass function *q*, then set $W^j := w(X^j) \geq 0$, where $w(\cdot)$ and *Q* satisfy $\mathbb{E}_Q\left[w(X)\right] = P(y)$. We relabel "cost" as "amount of success", set $S^j := s(X^j) \geq 0$ and specify upper and lower bounds, m_+ and m_-, on the number of samples. Each simulation brings a certain amount of success, and the simulation stops if a success threshold $\mathfrak{s}$ is reached or m_+ simulations have been performed. We summarise the relevant notation:

- $w^j = w(x^j)$: *weight*, or estimate of the quantity of interest, from the *j*th simulation.
- $s^j = s(x^j)$: *amount of success* for the *j*th simulation.
- $\mathfrak{s}$: required *total success*; simulation ends when the total success equals or exceeds $\mathfrak{s}$.
- m_-, m_+: the *minimum* and *maximum* number of simulations.

In general, $w(x^j) = f(y \mid x^j)\tilde{p}(x^j) / q(x^j)$, where $\tilde{p}$ is the prior mass/density for x^j; thus $\mathbb{E}_Q\left[w(X)\right] = P(y)$. When $Q = \tilde{P}$ and $f(y \mid x^j)$ is binary, then w^j is binary as in Algorithm 2.

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Algorithm 3 is valid for any non-negative $s(\cdot)$; we use $s(x) = f(y \mid x) / \sup_{x' \in \mathcal{X}} f(y \mid x')$, which is binary when *f* is binary and is justified more generally in Appendix D.

Algorithm 3 General One-step Frankenfilter

1: **procedure** FRANKENFILTERGENONE($\mathfrak{s}, m_-, m_+$) ▷ success threshold, min & max

2: Sample $x^{1:m_-}$ iid from *Q*; calculate weights, $w^{1:m_-}$ and measures of success $s^{1:m_-}$.

3: $m \leftarrow m_-$.

4: **while** $m < m_+$ **and** $\sum_{j=1}^{m} s^j < \mathfrak{s}$ **do**

5: $m \leftarrow m+1$.

6: Sample x^m from *Q*; calculate w^m and s^m.

7: **end while**

8: **if** $m = m_-$ **or** $\sum_{j=1}^{m} s^j < \mathfrak{s}$ **then return** $\hat{p} = \frac{1}{m} \sum_{j=1}^{m} w^j$.

9: **else return** $\hat{p} = \frac{1}{m-1} \sum_{j=1}^{m-1} w^j$.

10: **end if**

11: **end procedure**

On m_- : if $\operatorname{esssup} S^j \geq \mathfrak{s}$, setting $m_- \geq 1$ avoids an undefined estimate.

Proposition 1 . *The random variable* $\hat{P}$ *returned by Algorithm 3 satisfies*

$$\mathbb{E}\left[\hat{P}\right] = \mathbb{E}[W] = P(y).$$

Proof: Let *M*, W^j, S^j be the random variables corresponding to *m*, w^j, s^j.

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Let $K=0$ if $M=m_-$, $K=1$ if $M>m_-$ and $\sum_{j=1}^{m} S^j \geq \mathfrak{s}$ and $K=2$ if $\sum_{j=1}^{m_+} S^j < \mathfrak{s}$. The key concepts behind the proof of unbiasedness involve the exchangeability of the (W^j, S^j) that is induced by the stopping conditions:

• If $K=0$, $W^{1:M} \equiv W^{1:m_-}$ are exchangeable because they were all sampled at the same time and all contributed to the fact that $\sum_{j=1}^{m_-} S^j \geq \mathfrak{s}$.

• If $K=2$, $w^{1:M} \equiv W^{1:m_+}$ are exchangeable because the only additional information we have on the (S^j, W^j) pairs is that $\sum_{j=1}^{m_+} S^j < \mathfrak{s}$ (this implies that $\sum_{j=1}^{m} S^j < \mathfrak{s}$ for all $m < m_+$, so each of these earlier facts adds no extra information).

• If $K=1$ then the $W^{1:M-1}$ are exchangeable since all we know about the (S^j, W^j) pairs is that $\sum_{j=1}^{M-1} S^j < \mathfrak{s}$ and that adding S^M brings the total to at least $\mathfrak{s}$.

Let $I=1$ if $K=1$ and $I=0$ if $K \in \{0,2\}$. Now,

$$\mathbb{E}\left[\hat{P} \mid I=0\right] = \mathbb{E}\left[\frac{1}{M}\sum_{j=1}^{M} W^j \mid I=0\right] = \mathbb{E}\left[W^1 \mid I=0\right],$$

where the second equality follows from the exchangeability of the $\{(W^j, S^j)\}_{j=1}^{M}$ given $I=0$. Similarly,

$$\mathbb{E}\left[\hat{P} \mid I=1\right] = \mathbb{E}\left[\frac{1}{M-1}\sum_{j=1}^{M-1} W^j \mid I=1\right] = \mathbb{E}\left[W^1 \mid I=1\right].$$

Combining, and then applying the tower law:

$$\mathbb{E}\left[\hat{P}\right] = \mathbb{E}\left[\mathbb{E}\left[P \mid I\right]\right] = \mathbb{E}\left[\mathbb{E}\left[W^1 \mid I\right]\right] = \mathbb{E}\left[W^1\right]. \quad \square$$

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2.2 Partial and/or noisy observations

The Frankenfilter (Algorithm 4) extends Algorithm 3 to allow inference on multiple noisy and/or partial time-course observations. As in Algorithm 1, each sampled particle is propagated from one observation time to the next, either in a single jump or via a path, and we allow for resampling. The notation $X_{(t-1,t]}$ and $x_{(t-1,t]}$ denotes the random path of the particle over the interval $(t-1,t]$ and its realisation. The relevant pool of time-*t* ancestors depends on the outcome of the set of simulations from time $t-1$ to time *t* and is specified in lines 19 to 23. If $M = m_-$ ($K = 0$ in the proof of Proposition 1) or $\sum_{j=1}^{m_+} S^j < \mathfrak{s}$ ($K = 2$) then the pool consists of all *m* particles; otherwise ($K = 1$) it is all except the particle that caused the cumulative success to hit the threshold.

Suppose that $X_0 \sim p_0$ and $X_{(t-1,t]} \mid X_{t-1} = x \sim q(\cdot \mid x)$, an importance sampling (bridge) proposal. Let a_{t-1}^j be the time $t-1$ ancestor of particle x_t^j. In general

$$w^j = f(y_t \mid X_t = x_t^j)\, p(x_{(t-1,t]}^j \mid x_{t-1}^{a_{t-1}^j}) / q(x_{(t-1,t]}^j \mid x_{t-1}^{a_{t-1}^j}).$$

The algorithm fixes $\mathfrak{s}$ and $0 \le m_- < m_+$; if $\mathfrak{s} \le \max_{t=1,\ldots,T} \operatorname{esssup}(S_t^j)$, we insist $m_- \ge 1$.

Algorithm 4 General Frankenfilter

1: **procedure** FRANKENFILTER($\mathfrak{s}, m_-, m_+$) ▷ success threshold, min & max #trials

2: **for** *t* in 1: *T* **do**

3: **if** $t = 1$ **then**

4: Simulate $x_0^{1:m_-} \sim p_0$; set $a_{t-1}^{1:m_-} \leftarrow (1:m_-)$.

5: **else**

6: Sample $a_{t-1}^j \in \{1,\ldots,m_{\text{use}}\}$ with probability $\propto w_{\text{use}}^{1:m_{\text{use}}}$, $j = 1,\ldots,m_-$.

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7: **end if**

8: Sample $x_{(t-1,t)}^{j} \sim q(\cdot \mid x_{t-1}^{a_{t-1}^{j}})$; calculate w^{j} and s^{j}, $j = 1, \ldots, m_{-}$.

9: $m \leftarrow m_{-}$.

10: **while** $m < m_{+}$ **and** $\sum_{j=1}^{m} s^{j} < \mathfrak{s}$ **do**

11: $m \leftarrow m+1$.

12: **if** $t = 1$ **then**

13: Sample $x_{t-1}^{m} \sim p_0$; set $a_{t-1}^{m} = m$.

14: **else**

15: Sample $a_{t-1}^{m} \in \{1, \ldots, m_{\text{use}}\}$ with probability $\propto w_{\text{use}}^{1:m_{\text{use}}}$.

16: **end if**

17: Sample $x_{(t-1,t)}^{m} \sim q(\cdot \mid x_{t-1}^{a_{t-1}^{m}})$; calculate w^{m} and s^{m}.

18: **end while**

19: **if** $m = m_{-}$ **or** $\sum_{j=1}^{m_t} s^{j} < \mathfrak{s}$ **then**

20: $m_{\text{use}} \leftarrow m$

21: **else**

22: $m_{\text{use}} \leftarrow m-1$.

23: **end if**

24: $w_{\text{use}}^{1:m_{\text{use}}} \leftarrow w^{1:m_{use}}$; $\hat{p}_t \leftarrow \frac{1}{m_{\text{use}}} \sum_{j=1}^{m_{\text{use}}} w^{j}$

25: **end for**

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26: **return** $\hat{p} = \prod_{t=1}^{T} \hat{p}_t$.

27: **end procedure**

Theorem 1 . *The random variable* $\hat{P}$ *returned from Algorithm 4 satisfies:*

$$\mathbb{E}\left[\hat{P}(y_{1:T})\right] = P(y_{1:T}).$$

Theorem 1 is proved in Appendix A.

Generalisations: one might specify different budgets for each inter-observation interval, $\mathfrak{s}_{1:T}$, with $\mathfrak{s}_t$ based on a prior estimate of $p_t := P(y_t \mid y_{1:t-1})$, for example. This would still lead to an unbiased estimator of the likelihood. In Section 3, we show that with full exact observations it is reasonable to set $\mathfrak{s}_1 = \mathfrak{s}_2 = \ldots = \mathfrak{s}_T = \mathfrak{s}$. Simulations in Section 4 demonstrate that this is still a viable strategy, even with partial or noisy observations.

3 Tuning

Until recently, advice for tuning algorithms based on particle Metropolis–Hastings has been to choose the computational cost so that at some particular, representative parameter value, $\hat{\theta}$, $\mathrm{Var}\left[\log \hat{P}(Y_{1:T})\right] \approx 1\text{--}2$ Pitt et al. (2012); Sherlock et al. (2015); Doucet et al. (2015).

Sherlock (2024) points out that in the case where $\mathbb{P}\left(\hat{P} = 0\right) > 0$, $\mathrm{Var}\left[\log \hat{P}(y_{1:T})\right]$ is not even defined. This particular case occurs with exact or partial observations of an MJP where the likelihood is zero when the state does not fit with the observation. More generally, Sherlock (2024) shows that tuning so that the *relative variance*,

$$V_{\mathrm{rel}} := \frac{\mathrm{Var}\left[\hat{P}(y_{1:T})\right]}{\mathbb{E}\left[\hat{P}(y_{1:T})\right]^2} = \frac{1}{P(y_{1:T})^2}\mathrm{Var}\left[\hat{P}(y_{1:T})\right] \approx 1\text{--}2, \quad (1)$$

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or a little above 1, is a more robust approach as it can highlight potential large or infinite polynomial moments of $\hat{P}$. When $\hat{P}$ is well behaved and strictly positive, a first-order Taylor approximation shows that the two approaches are essentially equivalent.

All of the above advice assumes that $\mathrm{Var}\left[\hat{P}\right]$ or $\mathrm{Var}\left[\log\hat{P}\right]$ is inversely proportional to the computational cost. In the case of a standard particle filter, the variance is approximately inversely proportional to the number of particles, which is proportional to the cost. For the Alive particle filter, Taylor expanding (2) (see later) shows that the variance is approximately inversely proportional to $\mathfrak{s}$, which is proportional to the expected cost.

Achieving too small a variance simply increases the computational cost slightly; too large a variance can negatively impact the mixing of the algorithm, potentially substantially.

Increasing m_- can only lead to an estimator that is an average of more (or the same number) of terms, thereby reducing the Monte Carlo variance of $\hat{P}$ and making any recommended $\mathfrak{s}$ conservative. We will typically set m_+ sufficiently large that this upper threshold is met infrequently. Thus, some of our analysis is for the Alive particle filter, anticipating any resulting guidance will also apply to the Frankenfilter. We then examine the effect of m_+ on the overall variance and derive guidance on setting m_+.

3.1 Tuning for full, exact observations with binary weights

Initially, we allow for different success thresholds for each observation, $\mathfrak{s}_1,\ldots,\mathfrak{s}_T$. In the case of exact observations, the alive particle filter estimator of the transition probability from time $t-1$ to time *t* is

$$\hat{P}_t = \frac{\mathfrak{s}_t - 1}{M_t - 1},$$

where M_t is the number of attempts needed to obtain $\mathfrak{s}_t$ successes.

In the case of full, exact observations,

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$$V_{\text{rel}} = \frac{1}{\prod_{t=1}^{T} p_t^2} \mathbb{E}\left[\prod_{t=1}^{T} \hat{P}_t^2\right] - 1 = \left\{\prod_{t=1}^{T} \frac{1}{p_t^2} \mathbb{E}\left[\hat{P}_t^2\right]\right\} - 1.$$

The following result, proved in Appendix B, enables us to progress the analysis of V_{rel}.

Proposition 2 . *Let P be the random variable, with* $\mathbb{E}[P] = p$ *, returned by Algorithm 2 with* $m_+ = \infty$*, and define* $E_{\text{rel}} := \mathbb{E}[P^2] / p^2$*. For* $\mathfrak{s} = 2$*,*

$$E_{\text{rel}} = -\frac{\log p}{1-p},$$

for $\mathfrak{s} = 3$*,*

$$1 + \frac{1-p}{3} \leq E_{\text{rel}} = \frac{2}{1-p} + \frac{2p\log p}{(1-p)^2} \leq 2 - p,$$

and for $\mathfrak{s} \geq 4$*,*

$$1 + \frac{1-(1+\frac{2}{\mathfrak{s}-3})p}{\mathfrak{s}-2} < E_{\text{rel}} < 1 + \frac{1-p}{\mathfrak{s}-2}.$$

First, when $\mathfrak{s} = 2$, we cannot control E_{rel} as $p \downarrow 0$. For $\mathfrak{s} \geq 3$, however, E_{rel} is bounded, and for large $\mathfrak{s}$, it behaves as $1 + (1-p)/(\mathfrak{s}-2)$.

Using Proposition 2, then that each $\mathfrak{s}_t$ is typically large, and finally that in the cases where one might consider using the alive particle filter, p_t is typically small,

$$\left\{\prod_{t=1}^{T} \frac{1}{p_t^2} \mathbb{E}\left[\hat{P}_t^2\right]\right\} \approx \prod_{t=1}^{T} \left\{1 + \frac{1-p_t}{\mathfrak{s}_t - 2}\right\} \approx \exp\left(\sum_{t=1}^{T} \frac{1-p_t}{\mathfrak{s}_t - 2}\right) \approx \exp\left(\sum_{t=1}^{T} \frac{1}{\mathfrak{s}_t - 2}\right).$$

Fixing this, the expected cost is $\propto \sum_{t=1}^{T} \mathfrak{s}_t$, which is minimised by setting $\mathfrak{s}_1 = \ldots = \mathfrak{s}_T = \mathfrak{s}$. Then,

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$$V_{\text{rel}} \approx \exp\left(\frac{T}{\mathfrak{s}-2}\right)-1 \approx 1\text{--}2, \quad (2)$$

for good mixing. Note that, requiring $V_{\text{rel}} = e-1$ suggests setting $\mathfrak{s} = 2+T \approx T$.

It is worth emphasising that, unlike the case of the standard particle filter, this tuning advice does *not* require a choice of a representative parameter value. The guidance is valid whatever the system and whatever the true (small) transition probabilities are.

3.2 Tuning for partial, exact observations

We consider the alive particle filter in the case of partial, exact observations and multinomial resampling, and anticipate similar behaviour for the Frankenfilter provided m_+ is large.

Let the full state at time *t* be $X_t = (X_t^*, Z_t)$, where X_t^* is observed precisely and Z_t is unobserved. At time $t-1$ there are $\mathfrak{s}-1$ particles (with values $z_{t-1}^1, \ldots z_{t-1}^{\mathfrak{s}-1}$). We sample Z_{t-1} uniformly at random from these and propagate $X_{t-1} = (x_{t-1}^*, Z_{t-1})$ forwards until time *t*, registering the simulation as a success if $X_t^* = x_t^*$, and a failure otherwise. This is repeated until we achieve $\mathfrak{s}$ successes at time *t*. The first $\mathfrak{s}-1$ successes provide $\underline{z}_t := (z_t^1, \ldots, z_t^{\mathfrak{s}-1})$.

Define $p_t(z) = \mathbb{P}\left(X_t = x_t \mid X_{t-1} = (x_{t-1}^*, z)\right)$ and for $j = 1, 2, \ldots, \mathfrak{s}-1$, let $\bar{p}_t = \frac{1}{\mathfrak{s}-1}\sum_{j=1}^{\mathfrak{s}-1} p_t(z_{t-1}^j)$; for the random variable, we write $\bar{P}_t = \frac{1}{\mathfrak{s}-1}\sum_{j=1}^{\mathfrak{s}-1} p_t(Z_{t-1}^j)$.

For $t \in \{1, \ldots, T\}$, let $\mathcal{F}_t$ be the results of all simulations up to and including time *t* (the filtration at time *t*). All the relevant information from $\mathcal{F}_{t-1}$ that will inform the simulated values and estimated success probability at time *t*, as well as the evolution beyond time *t* is contained in $\underline{Z}_t$.

There are two special aspects of our scenario that facilitate helpful analytical insight.

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S1: Because we are choosing the ancestor of the *j*th simulation uniformly at random, when simulating $\hat{P}_t$, the distribution of the number of trials until $\mathfrak{s}$ successes is indistinguishable from the distribution of the number of trials if each were a simple Bernoulli trial with success probability $\bar{P}_t$.

S2: Conditional on $\mathcal{F}_{t-2}$, $\hat{P}_{t-1}$ and $(\bar{P}_t,\ldots,\bar{P}_T)$ are independent as the forward simulations from time $t-2$ to time $t-1$ are mutually independent. In particular, the number of failures, $F_{t-1} := |\{i : X_{t-1}^{*i} \neq x_{t-1}^{*}\}|$ is independent of the state values at the successes, and, therefore, of $\underline{z}_{t-1}$. Now, $\hat{P}_{t-1} = (\mathfrak{s}-1)/(\mathfrak{s}+F_{t-1}-1)$, whereas $\bar{P}_t$ is a deterministic function of $\underline{Z}_{t-1}$ and the subsequent evolution only depends on $\underline{Z}_{t-1}$, not F_{t-1}.

As before, V_{rel} is as defined in (1); now, however, we only know that $p = \mathbb{E}\left[\prod_{t=1}^{T} \hat{P}_t\right]$. Using S1 and Proposition 2 we have

$$\mathbb{E}\left[\hat{P}_t^2 \mid \mathcal{F}_{t-1}\right] = \mathbb{E}\left[\hat{P}_t^2 \mid \underline{Z}_{t-1}\right] = \mathbb{E}\left[\hat{P}_t^2 \mid \bar{P}_t\right] \leq \left(1+\frac{1}{\mathfrak{s}-2}\right)\bar{P}_t^2, \quad t=1,\ldots,T, \qquad (3)$$

and, again from Proposition 2, when $\mathfrak{s}$ is large and each p_t is small, we expect this bound to be close to an equality. Theorem 2, below, is proved in Appendix C.

Theorem 2 . *In the case of the alive particle filter on partial, exact observations with the same target number of successes, $\mathfrak{s}$, for each observation,*

$$V_{\text{rel}} \leq \exp\left\{\frac{T}{\mathfrak{s}-2}\right\}\frac{\mathbb{E}\left[\prod_{t=1}^{T}\bar{P}_t^2\right]}{\mathbb{E}\left[\prod_{t=1}^{T}\bar{P}_t\right]^2} - 1 = \exp\left\{\frac{T}{\mathfrak{s}-2}\right\}\frac{1}{P(y_{1:T})^2}\mathbb{E}\left[\prod_{t=1}^{T}\bar{P}_t^2\right] - 1. \qquad (4)$$

When $\mathfrak{s}$ is large, we expect (4) to be close to an equality. Rewriting this gives,

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$$V_{\text{rel}} \approx \exp\left\{\frac{T}{\mathfrak{s}-2}\right\} - 1 + \exp\left\{\frac{T}{\mathfrak{s}-2}\right\} \frac{\text{Var}\left[\prod_{t=1}^{T} \bar{P}_t\right]}{P(y_{1:T})^2}.$$

We see that for a given $\mathfrak{s}$, the relative variance is larger than in the case of exact observations. So, to achieve a given target relative variance, $\mathfrak{s}$ must increase. Since each $\bar{P}_t$ is an average of $\mathfrak{s}-1$ (albeit correlated) probabilities, when starting from $\mathfrak{s} \approx T$, we anticipate $\text{Var}\left[\prod_{t=1}^{T} \bar{P}_t\right]$ decreasing roughly in proportion to $1/\mathfrak{s}$ as $\mathfrak{s}$ increases. Appendix E provides further analysis in the case of very large $\mathfrak{s}$.

3.3 Choice of m_+

We now show that choosing m_+ such that $m_+ p = \kappa \mathfrak{s}$ for some moderately sized κ, such as $\kappa = 10$, is sufficient to control the additional extra variance that comes from the introduction of m_+ into the alive particle filter. We restrict attention to the most obvious application of Algorithm 3, where $s^j = w^j / c$, $j = 1, 2, \ldots$, for some $c > 0$. Propositions 3 and 4 are proved in Appendix B.

Proposition 3 . *Let $c > 0$ and let P be the random variable, with $\mathbb{E}[P] = p$, returned by Algorithm 3 in the special case that $S^j = \frac{1}{c} W^j$, $j = 1, 2, \ldots$. If $0 \le W^j \le w_*$, $j = 1, 2, \ldots$ and $c\mathfrak{s} > 2w_*$,*

$$\frac{1}{p^2}\mathbb{E}\left[P^2\right] \le 1 + \frac{w_*}{c\mathfrak{s} - 2w_*} + \frac{c\mathfrak{s}}{m_+ p}\mathbb{P}\left(\sum_{j=1}^{m_+} S^j < \mathfrak{s}\right).$$

The term $1 + w_* / (c\mathfrak{s} - 2w_*)$ is a slight loosening of the upper bound in Proposition 2, while $\mathbb{P}\left(\sum_{j=1}^{m_+} S^j < \mathfrak{s}\right)$ is the probability that the bound m_+ will make a difference to the output. The result gives an upper bound on the relative variance, generalising the approximation in (2):

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$$V_{\text{rel}} \leq \exp\left(\frac{Tw_*}{c\mathfrak{s} - 2w_*} + \sum_{t=1}^{T} \frac{c\mathfrak{s}}{m_+ p_t} \mathbb{P}\left(\sum_{j=1}^{m_+} S_t^j < \mathfrak{s} \right) \right) - 1. \quad (5)$$

The additional variance comes from the second term in the sum, which we, therefore, wish to keep small but not negligible, since, if it were negligible, the Frankenfilter would perform nearly as much work as the alive filter. Bernstein's inequality gives:

Proposition 4 . *If* $m_+ p = \kappa c\mathfrak{s}$ *for some* $\kappa \geq 7/4$ *then*

$$\mathbb{P}\left(\sum_{j=1}^{m_+} S^j < \mathfrak{s} \right) < \exp\left\{ -\frac{3\left(\kappa - \frac{7}{4}\right) c\mathfrak{s}}{8w_*} \right\}.$$

Combining with Proposition 3, we obtain

$$\frac{1}{p^2} \mathbb{E}\left[P^2 \right] < 1 + \frac{w_*}{c\mathfrak{s} - 2w_*} + \frac{1}{\kappa} \exp\left\{ -\frac{3(\kappa - \frac{7}{4}) c\mathfrak{s}}{8w_*} \right\}.$$

In the typical regime where $c\mathfrak{s} / w_* >> 1$, the bound on the probability that m_+ actually plays a role in the form of *P*, is controlled with relatively small values of κ , so this is sufficient to control the whole final term.

In practice, we do not know each p_t ; however, during a trial run of pseudo-marginal Metropolis–Hastings (PMMH), it is straightforward to output the fraction of times the Frankenfilter finishes sampling without achieving total success $\mathfrak{s}$ (see Table 5, for example); m_+ can be set so that this is small but not negligible. Alternatively, a very approximate lower bound on each p_t could be used to set a reasonable m_+ directly; in other parts of Section 4 we use a preliminary run of the alive particle filter at the true parameter value.

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4 Simulations

In this section we empirically validate the tuning advice of Section 3 and evaluate the performance of the Frankenfilter inside PMMH by considering five applications based upon MJPs, their discretisation and their approximation through a stochastic differential equation (SDE).

We allow the index set, $\mathcal{T}$, of X_t to be discrete or continuous. To fix additional notation to that in Section 1, for MJPs, $\mathcal{X} \subseteq \mathbb{Z}^{d_x}$ and (typically) $\mathcal{Y} \subseteq \mathbb{Z}^{d_y}$; for SDEs, $\mathcal{X} \subseteq \mathbb{R}^{d_x}$ and (typically) $\mathcal{Y} \subseteq \mathbb{R}^{d_y}$. In our MJP examples, $y_i = F^\top x_{t_i}$ for some constant $d_x \times d_y$ matrix F and in our SDE example, $Y_i \sim \mathsf{Gamma}(\alpha, \alpha / x_{t_i})$ for some $\alpha > 0$. We consider three specific Markov process models as follows. In the applications in Sections 4.1 and 4.5, the process $(X_t)_{t \geq 0}$ is modelled as an MJP. Given the state of the system at time t as x_t, the process transitions to time $t+\tau$ via

$$X_{t+\tau} = x_t + \sum_{i=1}^{r} S^i \mathcal{P}_i\left(\int_t^{t+\tau} h_i(x_{t'}, \theta) dt'\right), \quad (6)$$

where $\mathcal{P}_i(t)$, $i = 1, \ldots, r$ are independent, unit rate Poisson processes, S^i is the ith row of a $d_x \times r$ scaling matrix S whose elements are non-negative integers and the hazard function $h(x_t) := (h_1(x_t, \theta), \ldots, h_r(x_t, \theta))^\top$ is a vector of event rates or hazards parameterised by θ. MJPs are routinely used to model reaction networks consisting of r reaction channels; in this case, X_t represents a vector of specie counts, θ is a vector of reaction rate constants and S is known as the stoichiometry matrix, which describes the effect of each reaction on each component of X_t (see e.g. Wilkinson, 2018). Exact simulation of the MJP is straightforward via Gillespie's direct method (Gillespie, 1977). For the applications in Sections 4.2 and 4.3 where simulating the MJP exactly is likely to be computationally preclusive, we consider a tau-leap approximation (Gillespie, 2001). In this case, the state x_t is assumed constant over the time interval $[t, t+\tau)$ so that (6) becomes

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$$X_{t+\tau} = x_t + \sum_{i=1}^{r} S^i \mathcal{P}_i\left(h_i(x_t,\theta)\tau\right). \qquad (7)$$

Tau-leaping permits an adaptive time-step τ (see e.g. Cao et al., 2007). In what follows, for simplicity, we assume a constant τ. In Section 4.4, $(X_t)_{t\geq 0}$ is the solution of a stochastic differential equation (SDE),

$$\mathrm{d}X_t = a(X_t)\mathrm{d}t + b(X_t)\mathrm{d}W_t, \;\; X_0 = x_0,$$

where $a:\mathbb{R}^{d_x} \to \mathbb{R}^{d_x}$ is the drift, $b:\mathbb{R}^{d_x} \to \mathbb{R}^{d_x \times d_x}$ is the volatility and W_t is a d_x-dimensional vector of independent Brownian motions.

For PMMH runs, all filters and simulators were coded in `C++` and implemented in R via the package `Rcpp`. All runs were performed on a desktop computer with an Intel Core i7-10700 processor and a 2.90GHz clock speed. For PMMH, a Gaussian random walk proposal was applied to the natural logarithm of the parameter (vector), with innovation variance $\gamma \mathrm{Var}\left[\log\theta \mid y_{1:n}\right]$, and the scaling γ chosen to meet a desired acceptance rate (see Table 1 of Schmon et al., 2021). Code for the SDE experiments was written in R and run on an Intel Core Ultra 5 235U (up to 4.9 GHz). Code to implement the Frankenfilter and competing methods for each application is available from https://github.com/AndyGolightly/Frankenfilter .

Throughout, we abbreviate the bootstrap particle filter to BSPF, the Frankenfilter to FF and the alive particle filter to APF. When tabulating results, for the BSPF, m_+ represents the actual number of particles used and for the APF it represents the hard threshold, after which the algorithm is terminated and returns an estimated likelihood of 0. Effective sample size is abbreviated to ESS and effective samples per second to ESS/s.

4.1 Pure death process

We consider an MJP representation of a pure death process with hazard $h(x_t,\theta) = \theta x_t$ corresponding to a single reaction channel of the form $\mathcal{X} \to \varnothing$, with stoichiometry matrix

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$S = -1$. For the model, the transition probability of X_t (given the initial state x_0) is tractable and given by

$$\mathbb{P}\left(X_t = x\right) = \mathsf{Bin}\left(x; x_0, e^{-\theta t}\right).$$

We simulated two synthetic data sets on the time interval [0, 50] with $x_0 = 100$ and $\theta = \theta_{\text{true}} = 0.01$. The first data set, denoted D50, consists of observations at integer times. The second data set, denoted D50mod, is as D50, albeit with the last two observations replaced by the lower 0.01% quantiles of $X_{49} \mid X_{48}$ and $X_{50} \mid X_{49}$ respectively so $(x_{48}, x_{49}, x_{50}) = (58, 57, 57)$ rather than $(58, 53, 48)$. We assumed that $\theta \sim \mathsf{Gamma}(10, 1000)$ *a priori* and ran PMMH for $50K$ iterations with: the BSPF, FF and APF using the settings in Table 1. The number of particles for the BSPF (m_+) was chosen by following the practical advice of Sherlock et al. (2015). For the FF, we set $m_- = 0$ and chose m_+ to be the number of particles used in the BSPF. To assess the posterior accuracy of each filter-driven PMMH scheme, we also ran a Metropolis-Hastings (MH) scheme using the *exact* likelihood, evaluated using the tractable transition probabilities.

It is evident from Table 1 that for D50, there is little difference in overall efficiency between PMMH schemes, as measured by effective sample size per second (ESS/s). FF outperforms BSPF and APF by factors of 2 and 1.3-2, respectively. However, failure to choose a sufficiently large number of simulations m_+ in APF results in biased inferences for θ, due to the use of realisations of a biased estimator of likelihood in PMMH (see Figure 1 demonstrating this bias at the ground truth θ). A $3\hat{\sigma}/\sqrt{\text{ESS}}$ interval for the posterior expectation of $\theta / \theta_{\text{true}}$ using the direct method is $(1.083, 1.093)$ which does not overlap with the same interval under APF, even for $m^+ = 10^4$. Both BSPF and FF on the other hand, give inferences consistent with the direct method. For D50mod, the outlying final two observations have transition probabilities (likelihoods), at the ground truth θ, of 2.3×10^{-4} and 1.5×10^{-4}, necessitating a much larger value of m_+ for BSPF, and, in order to obtain accurate inferences, for APF. Consequently, FF outperforms BSPF and APF by factors of 10 and 6-30 respectively. The particularly small ESS obtained for APF with $m_+ = 10^4$ was due to the failure of the algorithm to generate a non-zero

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estimate of the likelihood for most iterations. Whereas $3\hat{\sigma}/\sqrt{ESS}$ intervals for the posterior expectation using BSPF and FF contain the estimate from the direct method, the interval from APF with $m_+ = 10^6$ only just intersects the interval from the direct method, and the interval using the APF with $m_+ = 10^4$ does not intersect it at all.

4.2 Protein dimerisation

A simple model of protein dimerisation involves two reactions of the form $2\mathcal{P} \to \mathcal{P}_2$ and $\mathcal{P}_2 \to 2\mathcal{P}$, where $\mathcal{P}$ represents protein and $\mathcal{P}_2$ represents dimers of the protein. The stoichiometry matrix and hazard vector are given by

$$S = \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix}, \quad h(x_t, \theta) = (\theta_1 x_{1,t}(x_{1,t} - 1)/2, \theta_2 x_{2,t})^\top$$

where $x_{1,t}$ and $x_{2,t}$ denote the respective numbers of proteins and dimers, at time *t*. Using the MJP representation (6), we simulated several data sets using the settings in Table 2; since $x_1 + 2x_2$ is conserved, species numbers in the P*0b data sets are typically ten times those in the P*0a data sets, and so transition probabilities are typically reduced by a factor of 10. In what follows, inference is performed on the tau-leaping model in (7) with $\tau = 0.1$.

We used the Frankenfilter inside PMMH, with $m_- = 0$. Following the advice in Section 3.3, we set $m_+ = 10\,\mathfrak{s}/\min_j \hat{P}_j$, where $\hat{P}_j$ is the estimated transition probability at time *j*, obtained by running the alive particle filter with θ fixed at the ground truth and $\mathfrak{s} = 5T$. Hence, m_+ is chosen conservatively, so that the tuning advice for $\mathfrak{s}$ in Section 3.1 should be appropriate. We specified $\theta_1 \sim \mathsf{Gamma}(2, 500)$ and $\theta_2 \sim \mathsf{Gamma}(2, 2)$ *a priori* and ran PMMH for $50K$ iterations for each data set. We empirically validate the tuning advice of Section 3.1 by reporting overall efficiency (ESS/s) against various choices of the required total success $\mathfrak{s}$ in Figure 2. Since there are two parameter chains, we take ESS as multivariate ESS (see e.g. Vats et al., 2019). Setting $\mathfrak{s} = T$ is clearly a viable strategy, with this choice giving optimal or close to optimal (at least 85% of maximum efficiency) across all data sets. Adopting a slightly more

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conservative strategy by requiring $V_{rel} = 1$ suggests $\mathfrak{s} = 16, 45$ and 74 for data sets with $T = 10, 30$ and 50 observations. For data sets P10a, P30a and P50a, this would result in efficiency factors (efficiencies relative to the maximum obtained across the different $\mathfrak{s}$ values) of 77%, 79% and 71%, respectively. For data sets P10b, P30b and P50b, the relative efficiency factors are 91%, 87% and 89%, respectively. Data sets P10a, P30a and P50a have relatively large typical transition probabilities at the true θ. With the approximation of $\sum_{t=1}^{T} 1 - p_t \approx T$ made in deriving the tuning advice, we might expect the optimal choice of $\mathfrak{s}$ to be slightly lower than *T*; however, this is only clearly apparent for the P50a data set, where the posterior for θ is tighter about the truth.

4.3 Lotka–Volterra tau leap

The Lotka-Volterra model of predator-prey interaction is routinely used to validate and compare inference schemes (see e.g. Boys et al., 2008). The associated reaction network for prey ($\mathcal{X}_1$) and predators ($\mathcal{X}_2$) has three reactions of the form $\mathcal{X}_1 \to 2\mathcal{X}_1$, $\mathcal{X}_1 + \mathcal{X}_2 \to 2\mathcal{X}_2$ and $\mathcal{X}_2 \to \varnothing$, representing prey reproduction, prey death / predator reproduction, and predator death. The stoichiometry matrix and hazard vector are given by

$$S = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad h(x_t, \theta) = (\theta_1 x_{1,t}, \theta_2 x_{1,t} x_{2,t}, \theta_3 x_{2,t})^\top$$

where $x_{1,t}$ and $x_{2,t}$ denote the respective numbers of prey and predators, at time *t*. Using the MJP representation (6), we simulated four data sets using the settings in Table 3. In what follows, we take the tau-leaping model in (7) with $\tau = 0.1$ as the inferential model.

For each data set typical transition probabilities are $\mathcal{O}(10^{-3})$ at the ground truth parameter values (and likely to be considerably smaller as θ deviates from the ground truth). Consequently, implementing the vanilla Frankenfilter (with particle trajectories generated via forward simulation) is computationally prohibitive. Therefore, for each inter-observation interval

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$(t-\Delta t, t)$, we generate trajectories $x^j_{(t-\Delta t,t)}$ using the conditioned hazard of Golightly and Wilkinson (2015). For $s \in [t-\Delta t, t)$, the conditioned hazard takes the form

$$h(x_s, \theta \mid y_t) = h(x_s, \theta) + H(x_s, \theta) S^\top F \{F^\top S H(x_s, \theta) S^\top F (t-s)\}^{-1} F^\top \{x_t - x_s - S h(x_s, \theta)(t-s)\}$$

where $y_t = F^\top x_t$, $H(x_s, \theta) = \text{diag}\{h(x_s, \theta)\}$ and $F = I_2$ in the case of full observations, and $F = (1,0)^\top$ in the case of partial observations (of prey values only). Hence, the weight function takes the form

$$w(x^j_{(t-\Delta t,t)}) = I(F' x_t = F' x_t^j) \prod_{k=1}^{N} \prod_{i=1}^{3} \frac{\text{Po}\left(\Delta n_{i,t-\Delta t+k\tau}; h(x_{t-\Delta t+(k-1)\tau}, \theta)\, \tau\right)}{\text{Po}\left(\Delta n_{i,t-\Delta t+k\tau}; h(x_{t-\Delta t+(k-1)\tau}, \theta \mid y_t)\, \tau\right)} \quad (8)$$

where $\text{Po}(x; \lambda)$ denotes the Poisson probability mass function with rate λ, $\Delta n_{i,s}$ denotes the number of reactions of type *i* in the interval $(s-\tau, s)$ and $N = \Delta t / \tau$.

In the following numerical experiments, we took an independent prior specification with $\theta_i \sim \text{Gamma}(1,1)$, $i = 1,2,3$ and set m_- and m_+ as in Section 4.2. To validate the tuning advice in Section 3, we ran PMMH for $50K$ iterations for each data set for various choices of the required total success $\mathfrak{s}$; Figure 3 reports overall efficiency (ESS/s) against $\mathfrak{s}$. For data sets with full observations, setting $\mathfrak{s} = T$ gives optimal overall efficiency for LV40 and close to optimal overall efficiency for LV20 (of those values of $\mathfrak{s}$ for which overall efficiency is reported). Requiring $V_{rel} = 1$ gives $\mathfrak{s}$ values of 31 and 60 for LV20 and LV40 respectively; these correspond to overall efficiencies of approximately 95% of maximum efficiency. For data sets LV20prey and LV40prey, which have observations on prey only, we additionally computed the value of $\mathfrak{s}$ that satisfies (4) for $V_{rel} = 1$. This was achieved with replicate runs of the Frankenfilter, with p_t approximated via the transition density under the chemical Langevin equation (CLE, see e.g. Wilkinson, 2018), and the expectations in (4) replaced by their sample equivalents. That is,

$$p_t \approx \text{N}\left(F^\top x_t; F^\top (x_{t-\Delta t} + \alpha(x_{t-\Delta t}, \theta)\Delta t), F^\top \beta(x_{t-\Delta t}, \theta) F \Delta t\right)$$

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where $\mathrm{N}(\cdot; m, V)$ denotes the Gaussian density with mean vector m and variance matrix V, and

$$\alpha(x,\theta) = Sh(x,\theta), \qquad \beta(x,\theta) = S\mathrm{diag}\{h(x,\theta)\}S^{\top}. \quad (9)$$

Figure 3 (bottom panel) shows the resulting value of $\mathfrak{s}$ (as a vertical line). This value is broadly consistent with the largest overall efficiencies obtained by running the Frankenfilter inside PMMH for various choices of $\mathfrak{s}$, achieving 97% of maximum efficiency for LV20prey and 98% for LV40prey.

Finally, in Table 4, we compare the overall efficiency of PMMH when using the bootstrap particle filter (BSPF) and Frankenfilter (FF). All filters used the weight function in (8), the optimal value of $\mathfrak{s}$ (FF) and m_+ (BSPF), and the PMMH scheme was initialised at an estimate of the posterior mean of $\log\theta$. For the data sets with full observations, FF outperforms BSPF in terms of overall efficiency by a factor of 1.9-2.6. This reduces to 1.7-2.4 for the data sets with partial observations. It is worth noting that these results are predicated on the availability of an initial PMMH run to find key quantities such as the estimated posterior mean and variance, for use in the main monitoring runs. In reality, "good" starting values for the PMMH chain may not be readily available. To compare the performance of FF and BSPF over hypothetical initial runs, we ran PMMH using LV20prey, with an initial value given by the ground truth parameter vector multiplied by 5. For FF, we fixed $\mathfrak{s} = 2T$ and $m^+ = 3\times10^5$. For BSPF, we set $m^+ = 3500$, which gave around a 50% chance of generating a non-zero likelihood estimate at the initial value of the chain. Using these settings and 50K iterations, overall efficiency scales as 1: 4 for BSPF versus FF. Intuitively, after burn-in, FF is able to use a lower computational effort to meet the desired number of matches.

4.4 Lotka–Volterra SDE

Here we compare the FF and BSPF for inference on an SDE that is observed with noise. Specifically, we consider the SDE for the Lotka–Volterra model where the drift and volatility are given through (9): $a(X_t) = \alpha(X_t)$ and $b(X_t) = \sqrt{\beta(X_t)}$; for the matrix square root, we used the

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transpose of the Cholesky decomposition. We employed the same time discretisation as for the tau-leap: $\delta = 0.1$.

We created a noisy version of each observation in the LV20 data set : for $i = 1, \ldots T = 20$ and $j = 1, 2$, $Y_{i,j} \mid X_{t_i,j} = x \sim \text{Gamma}(\alpha, \alpha / x)$, so that $\mathbb{E}\left[Y_{i,j} \mid X_{t_i,j}\right] = X_{t_i,j}$ and $\text{SD}\left[Y_{i,j} \mid X_{t_i,j}\right] = X_{t_i,j} / \sqrt{\alpha}$. We chose $\alpha = 50$ to represent relatively precise observations.

Since both the FF and BSPF simulate X_t from its prior, the particle weights are $w_i^j = f(y_i \mid x_{t_i}^j)$ where (with subscripts represent vector components here, only) $f(y \mid x) = \text{Gamma}(y_1; \alpha, \alpha / x_1)\text{Gamma}(y_2; \alpha, \alpha / x_2)$. We measure success by

$$s_i^j = \frac{f(y_i \mid x_{t_i}^j)}{\sup_{x \in \mathcal{X}} f(y_i \mid x)}. \quad (10)$$

One full unit of success is achieved when the state fits the observation as well as it can, and the poorer the fit the lower the amount of success. Appendix D explans why this is a natural measure.

We fixed $\mathfrak{s} = T = 20$ and investigated the stability of the variance and the increase in CPU time as the parameter vector fits less and less well with the data. Specifically, let θ_* be the true parameter vector, given in Table 3. We ran the particle filter $n_{\text{rep}} = 500$ times with parameter vector $\theta = \eta\theta_*$, with η ranging between 0.72 and 1.15. This was performed with $m_+ = 10^4$ and $m_+ = 10^3$, denoted as FF4 and FF3, respectively. The BSPF was run with $m = 120$ so that at the value of η that maximised the likelihood, the sample variance of the log likelihood under the BSPF matched those under the FF. That these variances are ≈ 1 suggests that setting $\mathfrak{s} = T$ may still be approximately valid for noisy observations when (10) quantifies success.

For each η, Table 5 shows the log likelihood, the sample variances of the n_{rep} log likelihood estimates under FF4, FF3 and the BSPF and the CPU time taken by each algorithm; all bootstrap runs took between 59.7 and 61.2 CPU seconds so individual times are not reported separately.

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For each observation, we also found the fraction of replicates where the target amount of success was not achieved; this provided, for each observation, an estimate of $\mathbb{P}(S < \mathfrak{s})$. Table 5 provides the maximum (over observations) of these, and for FF3, the average (over observations), which bounds the same under FF4.

Firstly, recall the mathematically optimal sample variances in Sherlock et al. (2015) and Doucet et al. (2015) for the case where the idealised Metropolis–Hastings chain would mix slowly: 3.28 and 2.82, respectively. However, both analyses assume that $\mathsf{Var}\left[\hat{\ell}\right]$ does not vary with θ; since this is not true in practice, and the inefficiency of the underlying chain increases exponentially with the variance, both suggest caution, setting $\mathsf{Var}\left[\hat{\ell}\right] \approx 1\text{--}2$ at a representative θ. Table 5 show that this caution is justified: as the log-likelihood decreases to around 16 below is maximum, $\mathsf{Var}\left[\hat{\ell}\right]$ increases by nearly two orders of magnitude for the BSPF , but by only one order of magnitude for the FF. Moreover, it is only when the log likelihood is 10 below its maximum that FF variance estimates are too large, whereas large variances occur for log-likelihood discrepancies of only 5 when the BSPF is employed.

The stability of the FF variance comes at an additional CPU cost whose size depends on m_+. The variances with FF3 are similar to those obtained from FF4, yet the more extreme computational costs are avoided, suggesting that $m_+ = 10^3$ might be preferable and a small but not negligible average failure fraction, rather than a small maximum fraction, might be a better tuning aim.

4.5 CWD in Mule deer

Finally, we consider a real data set consisting of the cumulative number of deaths of mule deer due to chronic wasting disease (CWD) observed at various times for two separate outbreaks, 1974-1985 and 1992-2001 (see e.g. Miller et al., 2006). The data have been analysed by Sun et al. (2015) and Drovandi and McCutchan (2016) among others; we follow the latter by treating each outbreak as independent and fitting the MJP representation of a susceptible, exposed, infected, removed (SEIR) model (with additional transitions for birth and natural death) to the full data set, denoted by $y = ([y^1_{1:n_1}]^\top, [y^2_{1:n_2}]^\top)^\top$. The associated list of reactions, hazards and

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stoichiometries is given in Table 6. Here, the per capita contact rate is β, and the average time between exposure and death due to CWD ($1/\mu$) is apportioned between exposure and infection via the parameter $0 \le \alpha \le 1$. The parameters *a* and *m* represent the number of deer added annually and the per-capita per year death date respectively; these are fixed at values included in the data set.

For the parameters of interest, $\theta = (\beta, \mu, \alpha)^\top$, we followed Drovandi and McCutchan (2016) by taking an independent prior specification with $\beta \sim \mathsf{Beta}(2,10)$, $\mu \sim \mathsf{Beta}(2,5)$ and $\alpha \sim \mathsf{U}(0,1)$. We fixed $E_1 = 0$ and specified initial distributions $S_1 \sim \mathsf{DiscreteU}(10,50)$, and $I_1 \sim \mathsf{DiscreteU}(0,20)$ for both outbreaks.

As with any real data application, the main monitoring run of PMMH with a Gaussian random walk proposal requires a sensible initial parameter value and innovation variance. These quantities can be obtained from the output of a pilot run of PMMH, which in turn requires a sensible m^+ for BSPF and additionally $\mathfrak{s}$ for FF. Therefore, using each of 5 values of θ sampled from the prior, we found values of m_+ that gave $\mathsf{Var}\left[\log \hat{P}(y)\right] \approx 1$ and selected the largest value, denoted m_+^0, for the pilot run. We repeated this process for FF with m_+ fixed at m_+^0 to find a suitable $\mathfrak{s}$, denoted $\mathfrak{s}^0$. Using these values, pilot runs of PMMH using both BSPF and FF were performed for 50*K* iterations, with the output used to estimate $\mathbb{E}[\theta \mid y]$ and $\mathsf{Var}[\log \theta \mid y]$. Finally, the main monitoring runs (50*K* iterations) of PMMH used values $m_+ = m_+^1$ and $\mathfrak{s} = \mathfrak{s}^1$ such that $\mathsf{Var}\left[\log \hat{P}(y)\right] \approx 1$ when running BSPF and FF with θ fixed at $\mathbb{E}[\theta \mid y]$.

Figure 4 shows marginal posterior densities for each parameter value, which are consistent across schemes and with the output reported in Drovandi and McCutchan (2016). The performance of BSPF and FF can found in Table 7. It is evident that, based on the main monitoring run alone, FF outperforms BSPF in terms of overall efficiency by a factor of around 1.6. When the computational cost of the pilot run is taken into account, relative efficiency increases to a factor of 4. That is, during the pilot run under FF and after burn-in, most sampled

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parameter values require many fewer simulations to obtain the desired number of matches, relative to the number of particles required by BSPF during its pilot run.

5 Discussion

We have introduced the Frankenfilter, a close-to-fully alive particle filter that adapts the number of simulations to the difficulty of the problem whilst employing a principled mechanism for bounding the total amount of compute.

In Section 3, we derived advice for choosing the total number of successes, $\mathfrak{s}$, required over each inter-observation interval; in the special case of T exact observations, this simply involves setting $\mathfrak{s} \approx T$. This advice was derived for the alive particle filter, but we also showed that choosing $m_+ = \kappa \mathfrak{s} / p$ for a moderately sized κ controls the impact of m_+ and m_- on the variance of the estimator. Section 4 validated our the advice on the Frankenfilter.

The literature backs up the intuition that our advice of choosing $\mathfrak{s} \approx T$ should also apply to PMMH using the alive filter with a large hard threshold. Drovandi et al. (2016) uses a hard threshold of 10^5 to perform inference via particle Metropolis–Hastings. Preliminary runs allow tuning parameters to be set to approximately optimise efficiency. Where the values of $\mathfrak{s}$ are reported, we can see that: in Experiment 1, $\mathfrak{s} = 101$ and $T \approx 135$ (before the observations are uninformative zeros), in Experiment 3, $\mathfrak{s} = 201$ and $T \approx 240$ and for Experiment 4, $\mathfrak{s} = 101$ and $T = 100$. Using the same hard threshold, Golightly and Sherlock (2019) examines a data set with $T = 7$ and finds $\mathfrak{s} = 8$ to be optimal.

We employed $s(x^j) = f(y \mid x^j) / \sup_{x \in \mathcal{X}} f(y \mid x)$, justified heuristically in Appendix D; however, the Frankenfilter is valid for any choice of $s(\cdot)$, including the constant function, whereupon (as pointed out by a reviewer) it reduces to the standard particle filter. Whilst our choice works well in the simulation studies, it may not be optimal: further work could investigate other possible success functions. Similarly, of our tuning results, the advice derived from Proposition 2 applies, strictly, to full-vector, exact observations, Theorem 2 applies to partial, exact observations, and Propositions 3 and 4 require $s(\cdot) \propto w(\cdot)$. Simulation studies suggest that the advice from these,

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of taking $\mathfrak{s} \approx T$ and choosing m_+ such that it is reached on a small but non-negligible fraction of occasions, is at least reasonable, but there is considerable scope for further investigation.

Comparisons in Section 4 demonstrated the bias of the alive particle filter with a hard threshold, even when that threshold is high, highlighting the need for the Frankenfilter. The studies tackled a variety of Markov process and filtering scenarios and demonstrated the greater robustness and efficiency of the Frankenfilter when compared with an equivalent standard filter with a fixed number of particles; in some cases the improvement in efficiency was of over an order of magnitude.

In Section 4.4, when $\max \bar{I}_{\mathrm{FF4}}$ is small, the second term in Proposition 3 is controlled, yet the variance under the Frankenfilter still increases as η varies. Whilst the Frankenfilter controls for the difficulty of simulating a good path, it does not control for the positive correlation between the averages of the weights from one observation to the next.

Consistent with the studies of del Moral et al. (2015) for the alive particle filter without any threshold, we have provided non-asymptotic bounds on the relative variance of the estimator of the likelihood under certain assumptions, since such quantities are central to the performance of the PMMH algorithm. Central limit theorems (CLTs) are in principle likely to hold under appropriate assumptions, although they are less directly relevant to the use of the algorithm in PMMH. For example, Le Gland and Oudjane (2006) prove a CLT for an estimator of the likelihood that is a biased but an asymptotically negligible perturbation of the estimator considered here, in the case where there is no hard thresholding. It is reasonable to expect that their result can be transferred, when both the hard threshold m_+ and $\mathfrak{s}$ diverge to ∞. We leave this to future work, as there are several technical details involved in such arguments, as one may deduce from Le Gland and Oudjane (2006).

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6 Supplementary material

The supplementary material contains proofs of Theorems 1 and 2 and Propositions 2, 3 and 4. It also describes the intuition between the choice of $w^j = f(y \mid x^j) / \sup_{x \in \mathcal{X}} f(y \mid x^j)$, additional analysis following on from Theorem 2, and additional discussion.

Data Availability Statement: all data used in this paper were simulated randomly from the models, as specified in Section 4.

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Conflict of Interest: The authors report there are no competing interests to declare.

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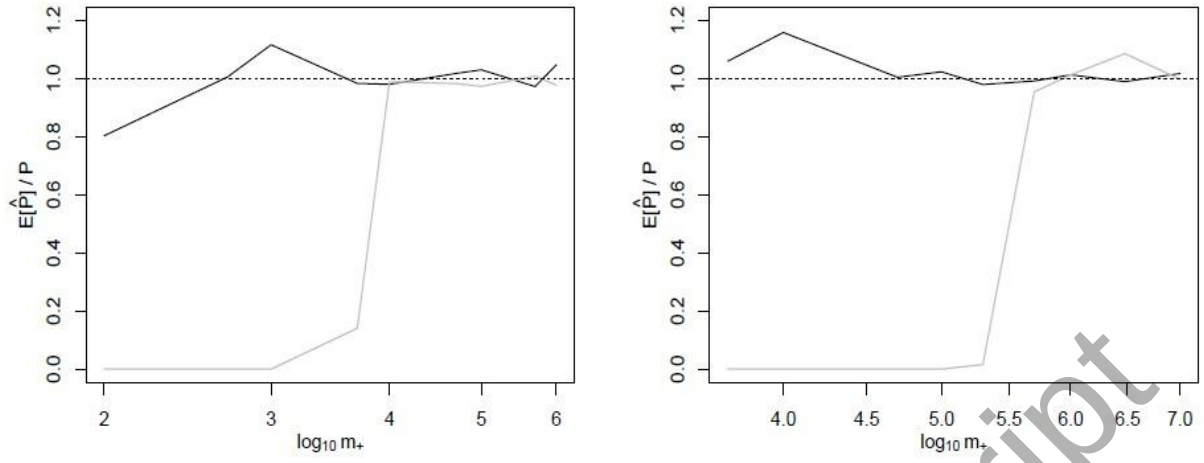


Figure 1: Death model. Ratio of the expectation of the estimator of the likelihood and exact likelihood at $\theta = 0.01$, based on the Frankenfilter (black lines) and alive particle filter (grey lines) using D50 (left panel) and D50mod (right panel).

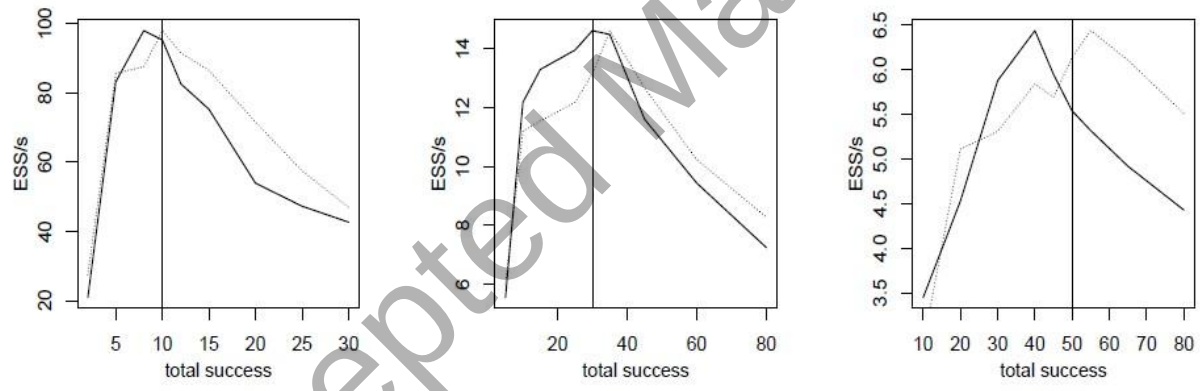


Figure 2: Protein dimerisation model. ESS/s versus required total success s, from the output of 50K iterations of PMMH, using data sets with 10 observations (left panel), 30 observations (middle panel), 50 observations (right panel). Solid lines correspond to data sets P10a, P30a and P50a whereas dotted lines correspond to data sets P10b, P30b and P50b (with ESS/s scaled to give the same maximum within observation regime). The value $s = T$ is indicated by a vertical line.

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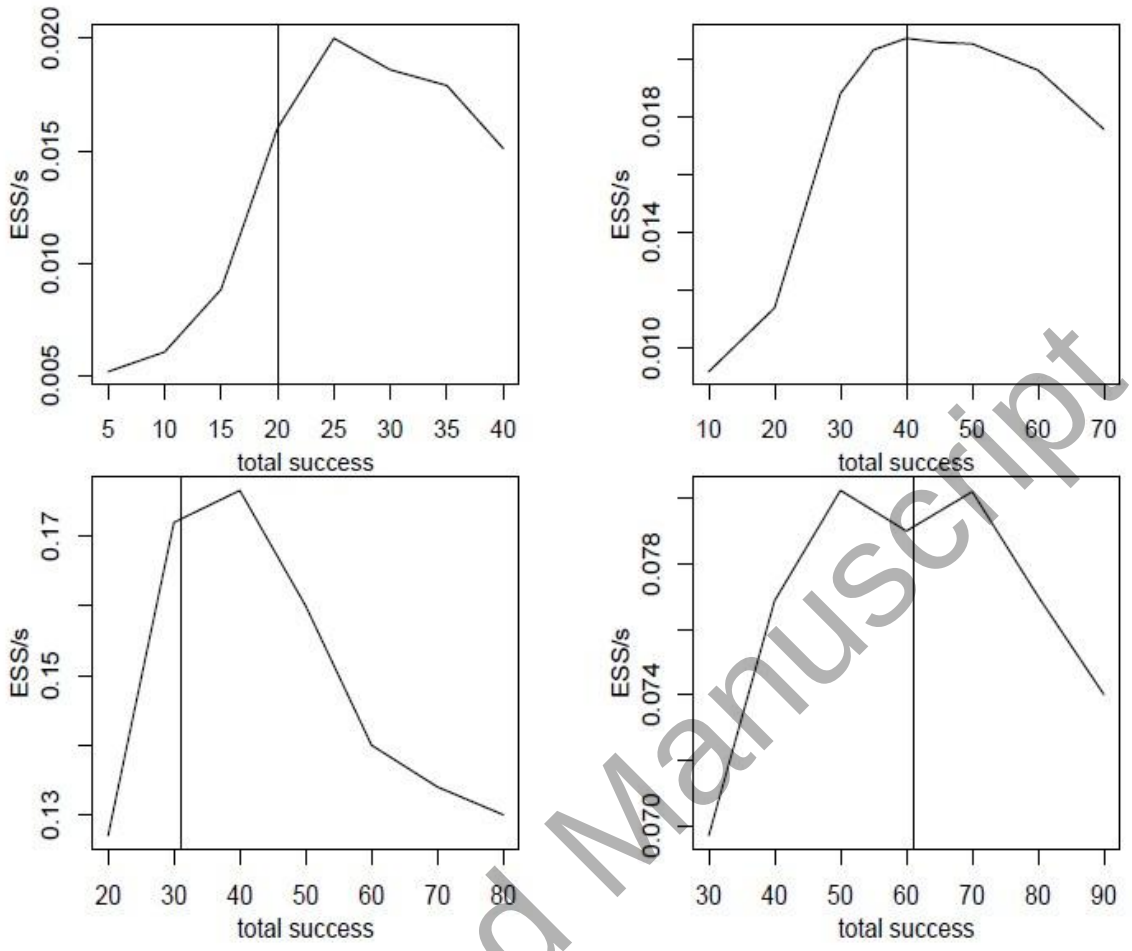


Figure 3: Lotka-Volterra tau leap. ESS/s versus required total success $\mathfrak{s}$, from the output of $50K$ iterations of PMMH, using data sets LV20 (top left), LV40 (top right), LV20prey (bottom left) and LV40prey (bottom right). Vertical lines show $\mathfrak{s} = T$ for the full observation data sets LV20 and LV40, and show the value of $\mathfrak{s}$ obtained by solving (4) with $V_{rel} = 1$ for the partial observation data sets LV20prey and LV40prey.

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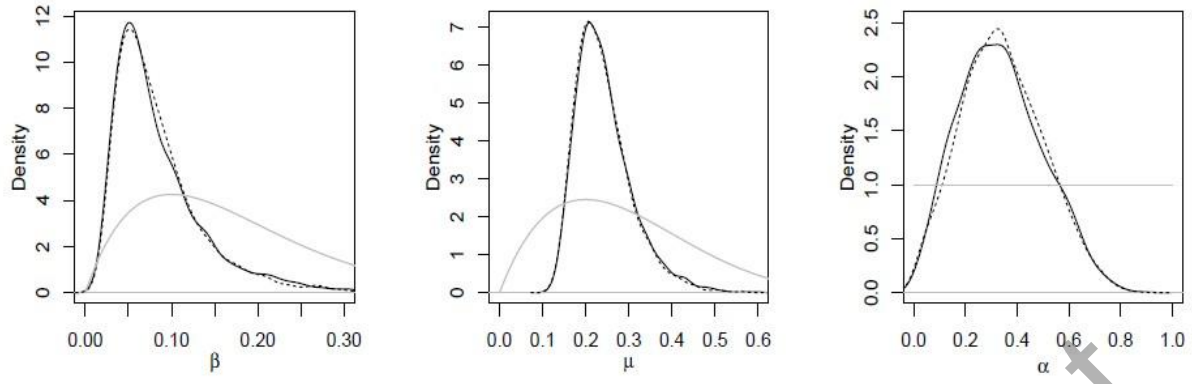


Figure 4: SEIR model. Marginal posterior densities of β, μ and α based on the output of PMMH with FF (solid black lines) and BSPF (dashed black lines). The prior distribution is shown in grey.

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Table 1: Death model. Filter used inside PMMH, required total success $\mathfrak{s}$, maximum number of simulations m_+, CPU time (in seconds), ESS, ESS/s, estimated posterior mean of $\theta / \theta_{\text{true}}$ (and standard deviation in parentheses) and standard error (SE) of this posterior mean. All results are based on $50K$ iterations of PMMH.

Data set	Filter	$\mathfrak{s}$	m_+	CPU (s)	ESS	ESS/s	Mean (SD)	SE
D50	Direct	-	-	12	8792	732.7	1.088 (0.149)	0.0016
	BSPF	-	400	527	2552	4.8	1.084 (0.148)	0.0029
	APF	50	400	226	1180	5.2	0.872 (0.200)	0.0058
		50	10^4	423	3263	7.7	1.107 (0.138)	0.0024
	FF	50	400	221	2227	10.1	1.085 (0.147)	0.0031
D50mod	Direct	-	-	12	9542	795.2	1.276 (0.163)	0.0017
	BSPF	-	10^4	11824	3415	0.3	1.280 (0.163)	0.0028
	APF	50	10^4	745	58	0.1	2.142 (0.081)	0.0107
		50	10^6	5899	3050	0.5	1.263 (0.160)	0.0029
	FF	50	10^4	799	2458	3.1	1.279 (0.160)	0.0032

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Table 2: Protein dimerisation model. Synthetic data set name, ground truth parameter values θ, initial condition x_0, inter-observation time Δt, final time T_{max}, number of observations T and average transition probability $p_t := \mathbb{P}\left(X_t = x_t \mid X_{t-1} = x_{t-1}\right)$ computed at the ground truth θ.

Data set	θ	x_0	Δt	T_{max}	T	$\frac{1}{T}\sum_{t=1}^{T} p_t$
P10a	(0.00332, 0.2)	(20, 1)	1	10	10	0.34
P10b	(0.00332, 0.2)	(200, 10)	1	10	10	0.07
P30a	(0.00332, 0.2)	(20, 1)	1	30	30	0.35
P30b	(0.00332, 0.2)	(200, 10)	1	30	30	0.08
P50a	(0.00332, 0.2)	(20, 1)	1	50	50	0.34
P50b	(0.00332, 0.2)	(200, 10)	1	50	50	0.07

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Table 3: Lotka-Volterra tau leap. Synthetic data set name, observed specie(s), ground truth parameter values θ, initial condition x_0, inter-observation time Δt, final time T_{max}, number of observations T and average transition probability, $p_t := \mathbb{P}\left(X_t = x_t \mid X_{t-\Delta t} = x_{t-\Delta t}\right)$ (for full observations) or $p_t := \mathbb{P}\left(Y_t = y_t \mid Y_{1:t-\Delta t} = y_{1:t-\Delta t}\right)$ (for partial observations) computed at the ground truth θ.

Data set	Obs	θ	x_0	Δt	T_{max}	T	$\frac{1}{T}\sum_{t=1}^{T} p_t$
LV20	$x_{1,t}, x_{2,t}$	(0.5, 0.0025, 0.3)	(50, 50)	1	20	20	0.00136
LV20prey	$x_{1,t}$	(0.5, 0.0025, 0.3)	(50, 50)	1	20	20	0.02836
LV40	$x_{1,t}, x_{2,t}$	(0.5, 0.0025, 0.3)	(50, 50)	0.5	20	40	0.00243
LV40prey	$x_{1,t}$	(0.5, 0.0025, 0.3)	(50, 50)	0.5	20	40	0.04339

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Table 4: Lotka-Volterra tau leap. Filter used inside PMMH, required total success $\mathfrak{s}$, maximum number of simulations m_+, CPU time (in seconds), effective sample size (ESS) and effective sample size per second (ESS/s). All results are based on 50K iterations of PMMH.

Data set	Filter	$\mathfrak{s}$	m_+	CPU (s)	ESS	ESS/s	Rel
LV20	BSPF	-	9500	244740	1893	0.0077	1.0
	FF	25	4×10^6	38163	770	0.0202	2.6
LV20prey	BSPF	-	900	36532	2680	0.0733	1.0
	FF	40	3×10^5	9840	1737	0.1766	2.4
LV40	BSPF	-	5400	143370	1548	0.0108	1.0
	FF	40	2×10^6	63680	1320	0.0207	1.9
LV40prey	BSPF	-	850	55007	2674	0.0486	1.0
	FF	70	2×10^5	23120	1854	0.0802	1.7

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Table 5: Lotka-Volterra SDE. The FF used $m_+ = 10^4$ (FF4) or $m_+ = 10^3$ (FF3); the BSPF used $m = 120$. For each η, 500 replicates of each filter were run. The table reports the log-likelihood (to 1dp), the sample variances of the log likelihood estimates for the three filters (V_{FF4}, V_{FF3} and V_{BS}). The CPU times (in seconds) for the 500 runs of FF4 and FF3, the minimum (over observations) of the fraction of replicates where $\mathfrak{s}$ was not achieved ($\max \bar{I}_{\mathrm{FF4}}$ and $\max \bar{I}_{\mathrm{FF3}}$), and the mean (over observations) of this fraction for FF3 ($\bar{\bar{I}}_{\mathrm{FF3}}$).

η	ℓ	V_{FF4}	V_{FF3}	V_{BS}	$\mathrm{CPU}_{\mathrm{FF4}}$	$\mathrm{CPU}_{\mathrm{FF3}}$	$\max \bar{I}_{\mathrm{FF4}}$	$\max \bar{I}_{\mathrm{FF3}}$	$\bar{\bar{I}}_{\mathrm{FF3}}$
0.72	-194.9	9.76	11.3	54.7	721.3	197.9	0.890	1.000	0.186
0.75	-189.3	6.74	8.05	25.0	356.2	146.9	0.482	0.996	0.106
0.80	-183.3	2.72	2.63	6.33	135.4	96.6	0.002	0.854	0.045
0.85	-179.8	1.13	1.14	1.97	78.0	77.4	0	0.094	0.007
0.90	-178.5	0.85	0.89	0.86	73.4	70.3	0	0.356	0.002
0.95	-179.2	1.01	0.95	1.09	87.2	72.2	0	0.758	0.040
1.00	-181.3	1.48	1.72	2.35	114.6	79.5	0.02	0.962	0.048
1.05	-184.7	3.33	3.06	5.15	166.6	98.5	0.010	0.996	0.057
1.10	-189.0	5.76	5.53	13.2	259.5	129.2	0.044	0.998	0.087
1.15	-194.5	9.47	11.3	35.4	537.7	185.6	0.240	1.000	0.127

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Table 6: SEIR model. Reaction, hazard, stoichiometry (effect on each specie S_t, I_t, E_t, R_t) and description.

Reaction	Hazard	Stoichiometry	Description
$\mathcal{R}_1 : \varnothing \to S$	a	$(1,0,0,0)^\top$	Birth
$\mathcal{R}_2 : S \to \varnothing$	ms_t	$(-1,0,0,0)^\top$	Susceptible death (natural)
$\mathcal{R}_3 : S + I \to E$	$\beta s_t i_t$	$(-1,1,0,0)^\top$	Contact
$\mathcal{R}_4 : E \to \varnothing$	me_t	$(0,-1,0,0)^\top$	Exposed death (natural)
$\mathcal{R}_5 : E \to I$	$\frac{\mu}{\alpha} e_t$	$(0,-1,1,0)^\top$	Infection
$\mathcal{R}_6 : I \to \varnothing$	mi_t	$(0,0,-1,0)^\top$	Infective death (natural)
$\mathcal{R}_7 : I \to R$	$\frac{\mu}{1-\alpha} i_t$	$(0,0,-1,1)^\top$	Infective death (CWD)

Table 7: SEIR model. Filter used inside PMMH, required total success $\mathfrak{s}^0$ ($\mathfrak{s}^1$), maximum number of simulations m_+^0 (m_+^1) and CPU time (in seconds) CPU^0 (CPU^1) used in the pilot run (main monitoring run), effective sample size (ESS) from the main monitoring run and effective sample size per total CPU time (ESS/TCPU).

Filter	$\mathfrak{s}^0$	m_+^0	CPU^0 (s)	$\mathfrak{s}^1$	m_+^1	CPU^1 (s)	ESS	ESS/TCPU	Rel
BSPF	-	6000	407628	-	1000	24777	1363	0.00315	1
FF	300	6000	89500	90	1000	15338	1328	0.01267	4

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